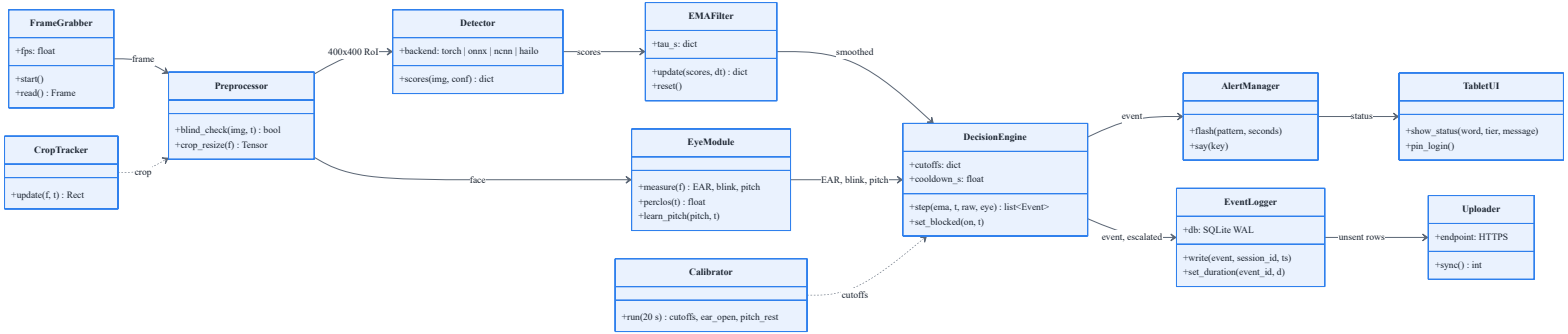

FrameGrabber
+fps: float
+start()
+read() : Frame
frame
Preprocessor
+blind_check(img, t) : bool
+crop_resize(f) : Tensor
400x400 RoI
Detector
+backend: torch | onnx | ncnn | hailo
+scores(img, conf) : dict
scores
EMAFilter
+tau_s: dict
+update(scores, dt) : dict
+reset()
smoothed
CropTracker
+update(f, t) : Rect
crop
face
EyeModule
+measure(f) : EAR, blink, pitch
+perclos(t) : float
+learn_pitch(pitch, t)
EAR, blink, pitch
DecisionEngine
+cutoffs: dict
+cooldown_s: float
+step(ema, t, raw, eye) : list<Event>
+set_blocked(on, t)
event
AlertManager
+flash(pattern, seconds)
+say(key)
status
TabletUI
+show_status(word, tier, message)
+pin_login()
event, escalated
EventLogger
+db: SQLite WAL
+write(event, session_id, ts)
+set_duration(event_id, d)
unsent rows
Uploader
+endpoint: HTTPS
+sync() : int
Calibrator
+run(20 s) : cutoffs, ear_open, pitch_rest
cutoffs